

Don't Trust All Your Neighbors! Analyzing Graph Attention Networks

CS 4782: Deep Learning · Cornell University · Spring 2026

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1 Introduction & Background

- Goal: given a citation network of papers, predict each paper's research topic using its text and citation links, knowing labels for only ~140 of ~2,700 papers
- GCNs** exploit this by aggregating neighbor features, but treating every neighbor equally lets a loosely related paper get the same weight as a highly relevant one
- GATs** fix this by learning a separate attention score per edge, so the model decides which neighbors to trust
- We reproduce Table 2: transductive node classification on **Cora** (ML papers, 7 topics) and **CiteSeer** (CS papers, 6 topics), averaged over 100 runs

Transductive			
Method	Cora	Citeseer	Pubmed
MLP	55.1%	46.5%	71.4%
MamIReg (Belkin et al., 2006)	59.5%	60.1%	70.7%
SemimEmb (Weston et al., 2012)	59.0%	59.6%	71.7%
LP (Zhu et al., 2003)	68.0%	45.3%	63.0%
DeepWalk (Perozzi et al., 2014)	67.2%	43.2%	65.3%
ICA (Lu & Getoor, 2003)	75.1%	69.1%	73.9%
Planetoid (Yang et al., 2016)	75.7%	64.7%	77.2%
Chebyshev (Defferrard et al., 2016)	81.2%	69.8%	74.4%
GCN (Kipf & Welling, 2017)	81.5%	70.3%	79.0%
MoNet (Monti et al., 2016)	81.7 ± 0.5%	—	78.8 ± 0.3%
GCN-64*	81.4 ± 0.5%	70.9 ± 0.5%	79.0 ± 0.3%
GAT (ours)	83.0 ± 0.7%	72.5 ± 0.7%	79.0 ± 0.3%

2 Methodology

Input Data

Datasets

Cora

CiteSeer

Loader

PyG Planetoid

NormalizeFeatures()

Training Splits

20
NODES

500
NODES

1000
NODES

Train

Val

Test

LAYER 1

GATConv

Heads

8

Feat/Head

8

Output

64-dim

Activation

ELU

Dropout

0.6

64-Dim Representation

LAYER 2

GATConv

Head

1

Output

num_classes

Activation

Log-Softmax

Dropout

0.6

Training Setup & Output

Optimizer

Adam · lr = 0.005

Weight Decay

5e-4 (λ = 0.0005)

Early Stop

patience = 100, val loss

Output

Log-Probabilities

3 Results

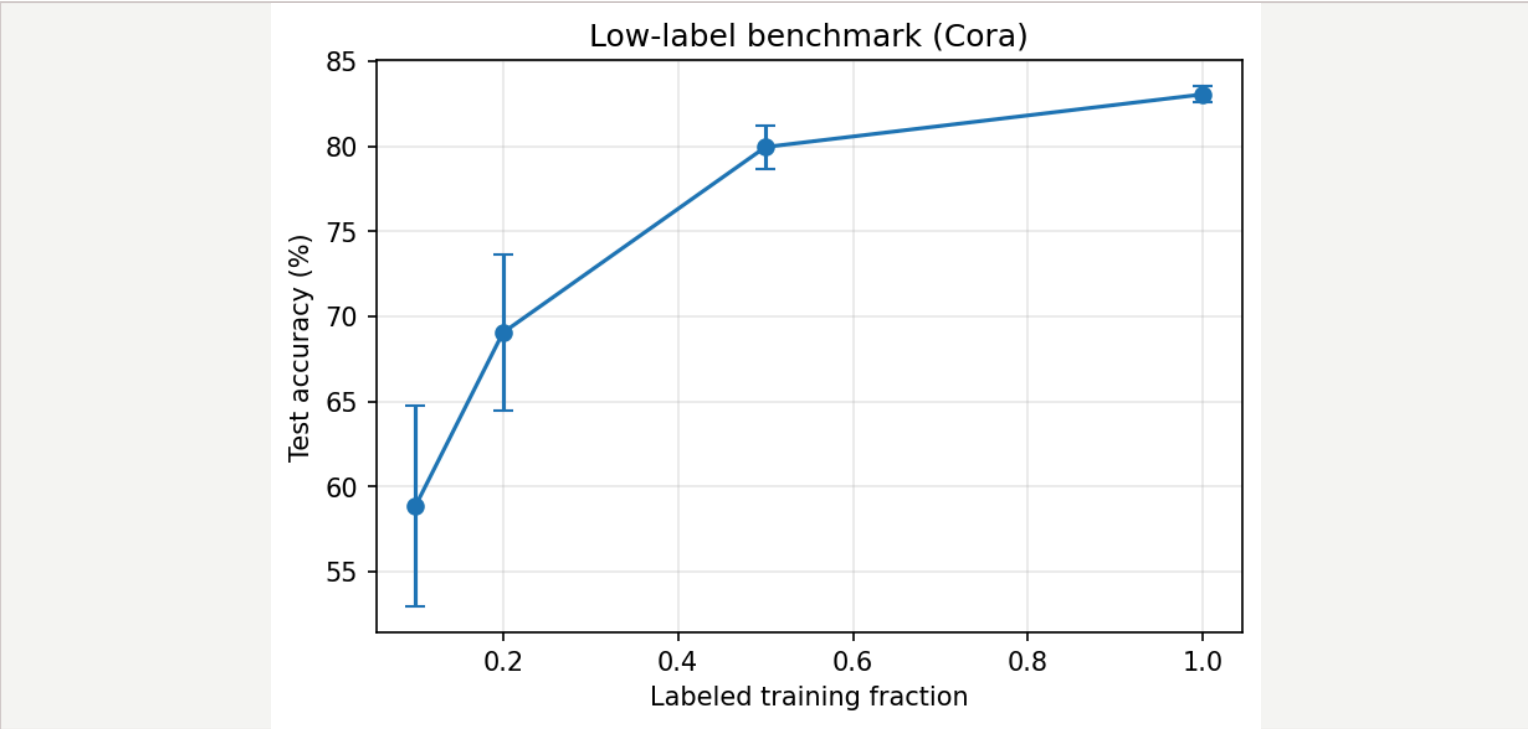
BASELINE REPRODUCTION

Dataset	Paper Target	Ours	Status
Cora	83.0 ± 0.7%	83.22 ± 0.41%	✓ Matched
CiteSeer	72.5 ± 0.7%	70.94 ± 0.52%	-1.56% †

† CiteSeer gap: early stopping monitored only validation loss, not loss + accuracy as in the original TF implementation.

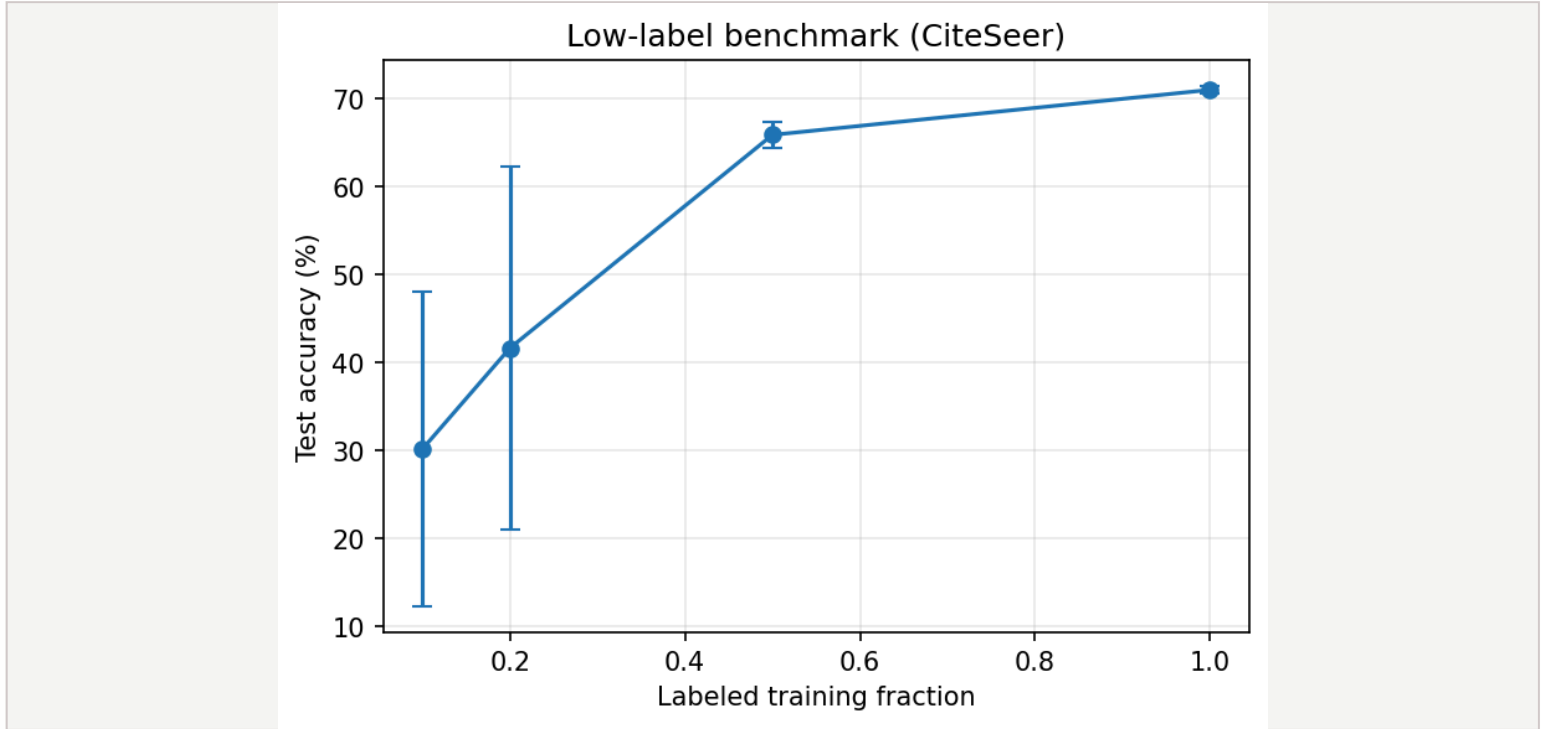
Cora

WHAT IF DATA ISN'T FULLY LABELLED?



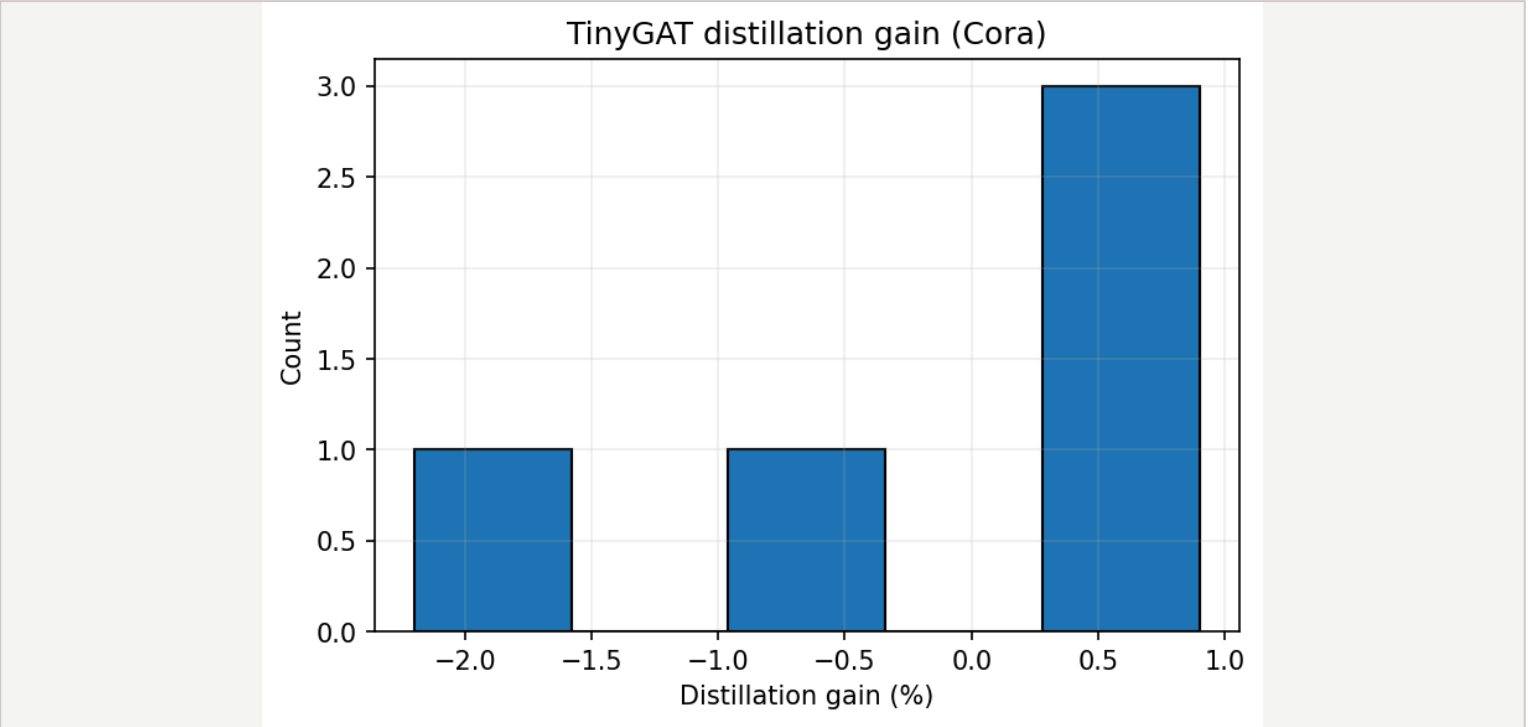
Accuracy 83% → 59% as labels shrink; std = 5.9% at 10%.

CiteSeer

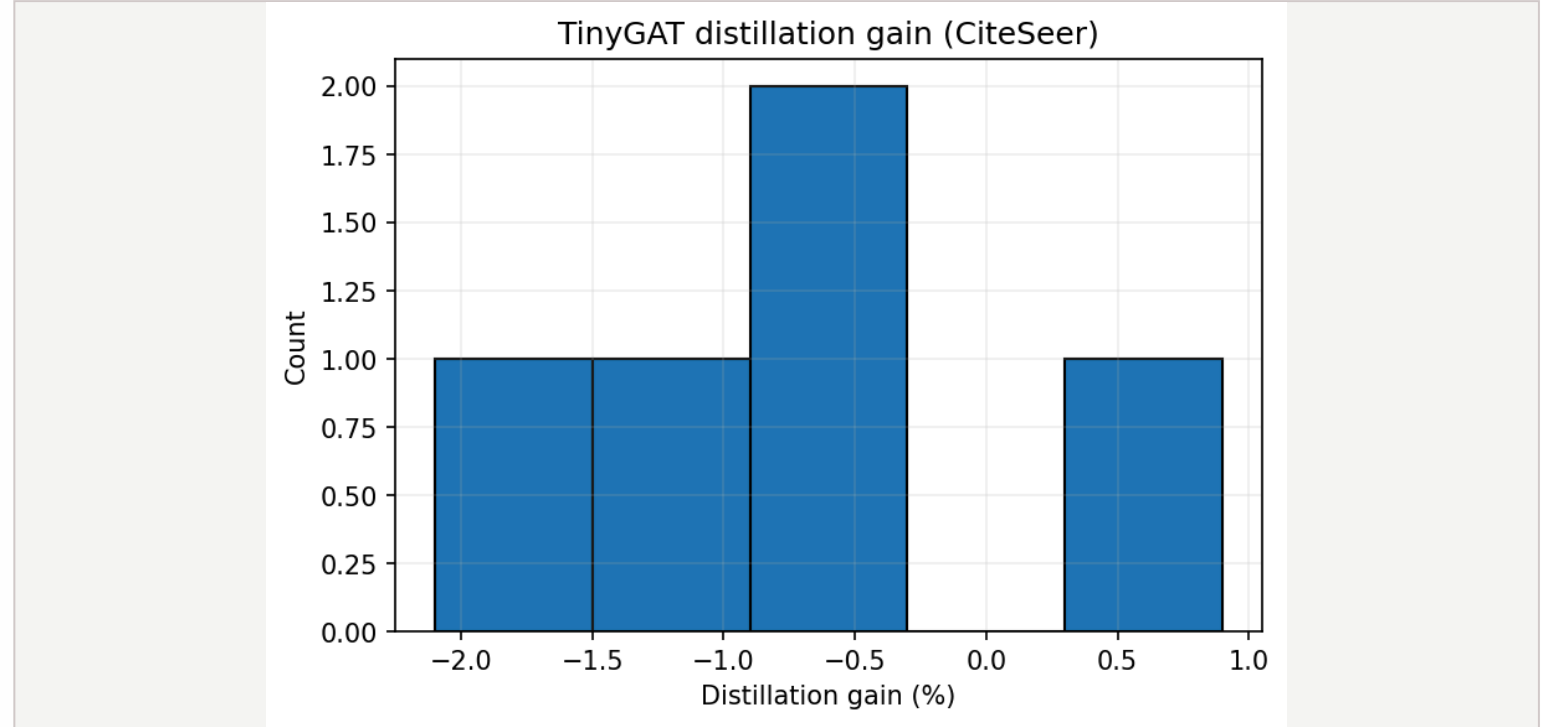


Accuracy 71% → 30% as labels shrink; high variance at 10%.

HOW DOES CHEAPER TRAINING TRADE OFF WITH ACCURACY?

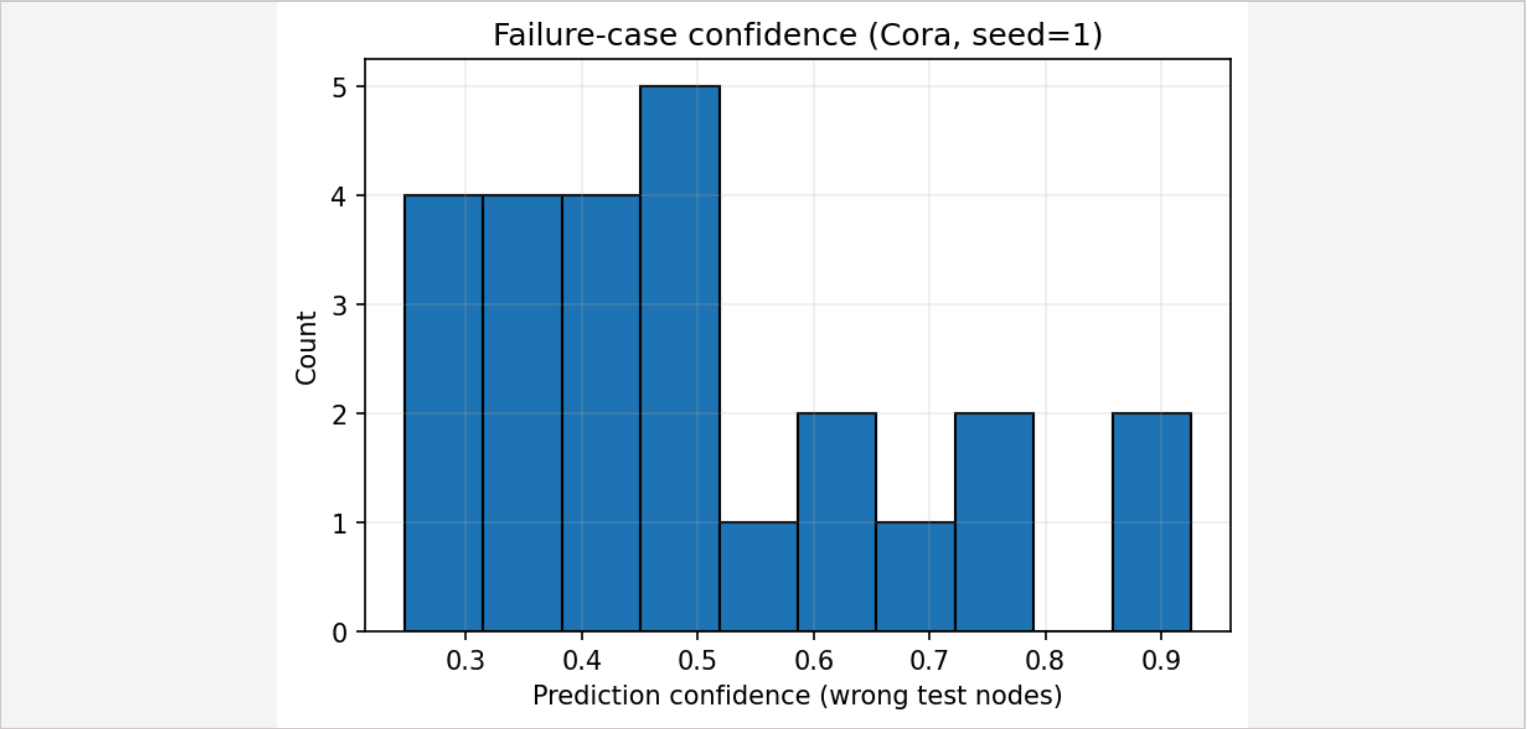


Mean gain = -0.24%; distillation gives minimal benefit.

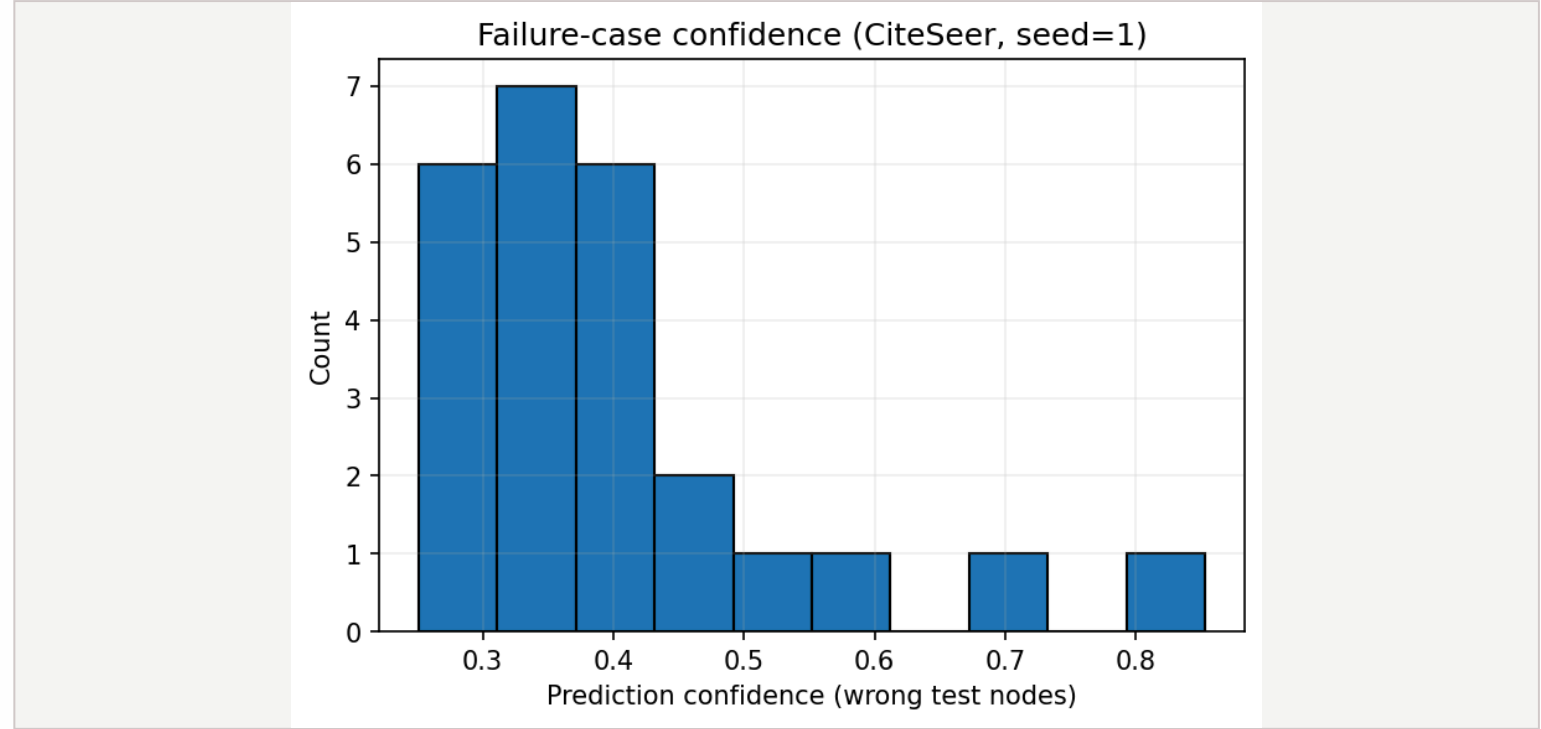


Distillation gain mostly negative; TinyGAT already near-ceiling.

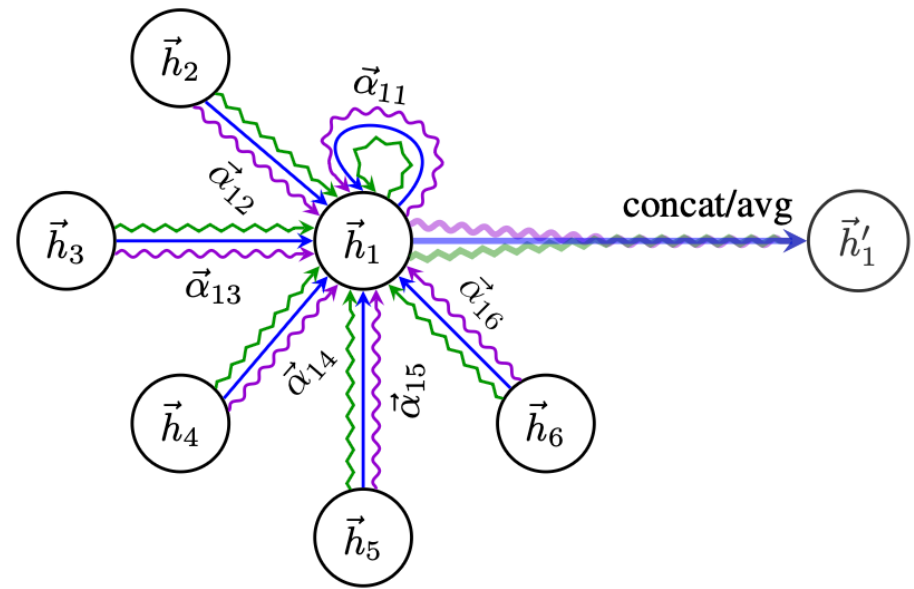
HOW OFTEN DO WE GET THINGS CONFIDENTLY WRONG?



Two high-confidence errors (92%, 93%) — attention on wrong-class neighbors.



Most failures are low-confidence (0.3–0.4); model more uncertain on CiteSeer.



$\vec{h}_j$ is the feature vector of neighbor j ; α_{1j} is the learned attention weight controlling how much node 1 attends to it. Each color is an independent attention head with its own weighting. The outputs are concatenated or averaged to form $\vec{h}_1'$, the updated node embedding.

4 Conclusion

- We were able to successfully **reproduce the paper's original result** on the Cora dataset
- GAT degrades sharply under label scarcity. Performance goes down significantly when labels drop from 55% to 20%, and an even sharper decrease when we go from 20% to 8%.
- Distillation did not provide many benefits on average, but TinyGAT already matched the teacher model very closely. We can get more efficiency without trading off on accuracy!
- When the model is confidently wrong, its nodes are from multiple classes! The model struggles to allocate attention accurately when a paper's citations come from many different fields.

5 Future Work

- Run extensions on PubMed and PPI datasets for **broader generalization**; PubMed tests transductive scaling, PPI tests the inductive setting the paper also addresses
- Explore attention mechanisms that can actively repel misleading neighbors rather than just reducing their weight.
- Explore why CiteSeer collapses at low labels, isolate whether it's feature sparsity, graph topology, or isolated nodes by ablating each factor

REF References

- Veličković et al. (2018). *Graph Attention Networks*. ICLR 2018.
- Fey & Lenssen (2019). *Fast Graph Representation Learning with PyTorch Geometric*. ICLR Workshop.
- Yang et al. (2016). *Revisiting Semi-Supervised Learning with Graph Embeddings*. ICML.